



Slides 5.13: Cost Per Wear and Quality Checks

Lesson 5.13, one day (day 133). Thirteen slides. All prices to be refreshed before teaching.

Slide 1: Do now

- A pair of jeans costs \$60.
- You wear them 200 times before they wear out.
- How much did each wearing cost you?
- Show the division.

Notes: two minutes. Take the answer and the division out loud. $\$60 \div 200 = \0.30 per wear. Insist on the unit. Then the hook: a \$15 shirt worn five times. Do that in your head and tell me which one was expensive.

Slide 2: Cost per wear

- **Cost per wear = price $\div$ the number of times worn**
- The answer is in **dollars per wear**.
- Price is what you pay once.
- Cost per wear is what the item actually costs you.

Slide 3: Worked problem (a), the jeans

- Price: \$60
- Wears: 200
- $\$60 \div 200 = \mathbf{\$0.30 \text{ per wear}}$

Slide 4: Worked problem (b), the shirt

- Price: \$15
- Wears: 5
- $\$15 \div 5 = \mathbf{\$3.00 \text{ per wear}}$
- Cheapest price on the slide. Most expensive to own.

Slide 5: Worked problem (c), the coat

- Price: \$90
- Worn 3 winters. How many wears is that?
- How many times do you wear a winter coat in one winter? _____
- $3 \times \text{_____} = \text{_____}$ wears
- $\$90 \div \text{_____} = \_____ per wear
- **The estimate is the argument.** You have to decide the number of wears, and then defend it.

Notes: let two students disagree about how many cold days there are. That disagreement is the standard (SUS 1. f), the cost and benefit of a decision). At 60 wears per winter the answer is \$0.50 per wear.

Slide 6: What makes a garment last, 1: fiber

- Cotton: comfortable, absorbent, wrinkles, shrinks
- Wool: warm even when damp, can felt with heat and agitation

- Polyester: resists wrinkles, dries fast, holds odor, melts under a hot iron
- Spandex or elastane: stretch and recovery, and it is the fiber that fails first
- A cotton and polyester blend survives washing better than either one alone.

Image: four fabric swatches labeled with their fiber.

Slide 7: What makes a garment last, 2: seam type

- **Single-needle:** one row of stitching. Opens first.
- **Double-needle:** two parallel rows. Holds.
- **Flat felled:** raw edges folded and wrapped inside, then stitched through. Strongest.
- **Serged edge:** thread looped over a raw edge. That is a finish, not a seam.

Notes: turn a pair of jeans inside out and find the flat felled side seam. Then find the serged edge inside a cheap shirt.

Image: four close-ups, one per seam type, from the inside of a garment.

Slide 8: What makes a garment last, 3: stitches per inch

- Put the ruler on a structural seam.
- Count the stitches in one inch.
- 8 or fewer is low.
- 10 to 12 holds.
- More stitches per inch means more thread holding the same inch of seam.

Image: a ruler along a seam with a bracket marking one inch.

Slide 9: What makes a garment last, 4: reinforcement at stress points

- A **stress point** is anywhere the body pulls on the garment.
- Pocket corners. Belt loops. Under the arm. The knee. Where a strap joins.
- Look for a **bar tack**: a short, dense line of stitches at that spot.
- A gusset, a doubled layer, a knee patch: all reinforcement.
- No reinforcement at a stress point is where the garment will fail.

Image: a bar tack at the corner of a jeans pocket, magnified.

Slide 10: What makes a garment last, 5: buttons and closures

- Four to six passes of thread through the holes.
- A **shank**: a little slack so the button sits above the fabric and the buttonhole has room.
- A knot on the wrong side that holds, with a tail, not cut flush.
- A zipper that runs smoothly both ways.

Notes: hold up a well-sewn button and a loose one. Do not sew today.

Image: a button with a visible shank next to a button lying flat with two loose threads.

Slide 11: Inspection checklist, five checks per garment

Check	Good
Fiber from the label	Percentages printed, a blend or a durable fiber
Seam type	Double-needle or flat felled on load-bearing seams
Stitches per inch	10 to 12
Reinforcement at a stress point	A bar tack, a gusset, or a doubled layer
Buttons and closures	Enough passes, a shank, a knot that holds

Notes: three minutes per station, one bell, three rotations. Refuse to say whether a garment is “good.” The checklist decides.

Slide 12: The trade-off

- Garment 2, the thin t-shirt: \$8
- Garment 1, the heavy sweatshirt: \$44
- **Person 1:** walks to school, wears it four days a week for two winters.
- **Person 2:** needs a shirt for one concert.
- Compute cost per wear for both garments, for both people.
- Then say who should buy which.

Notes: the point out loud: cost per wear only answers the question once you know who is wearing it and how often.

Slide 13: Exit card

- 3 things you would check on a garment before buying it
- 2 cost-per-wear numbers you computed today, with units
- 1 item in your own closet you now think was a good buy or a bad buy, and why in six words

Notes: three students read. The “needs more” pile decides whether Lesson 5.14 Day 2 opens with a 90-second cost-per-wear review before the graded task.